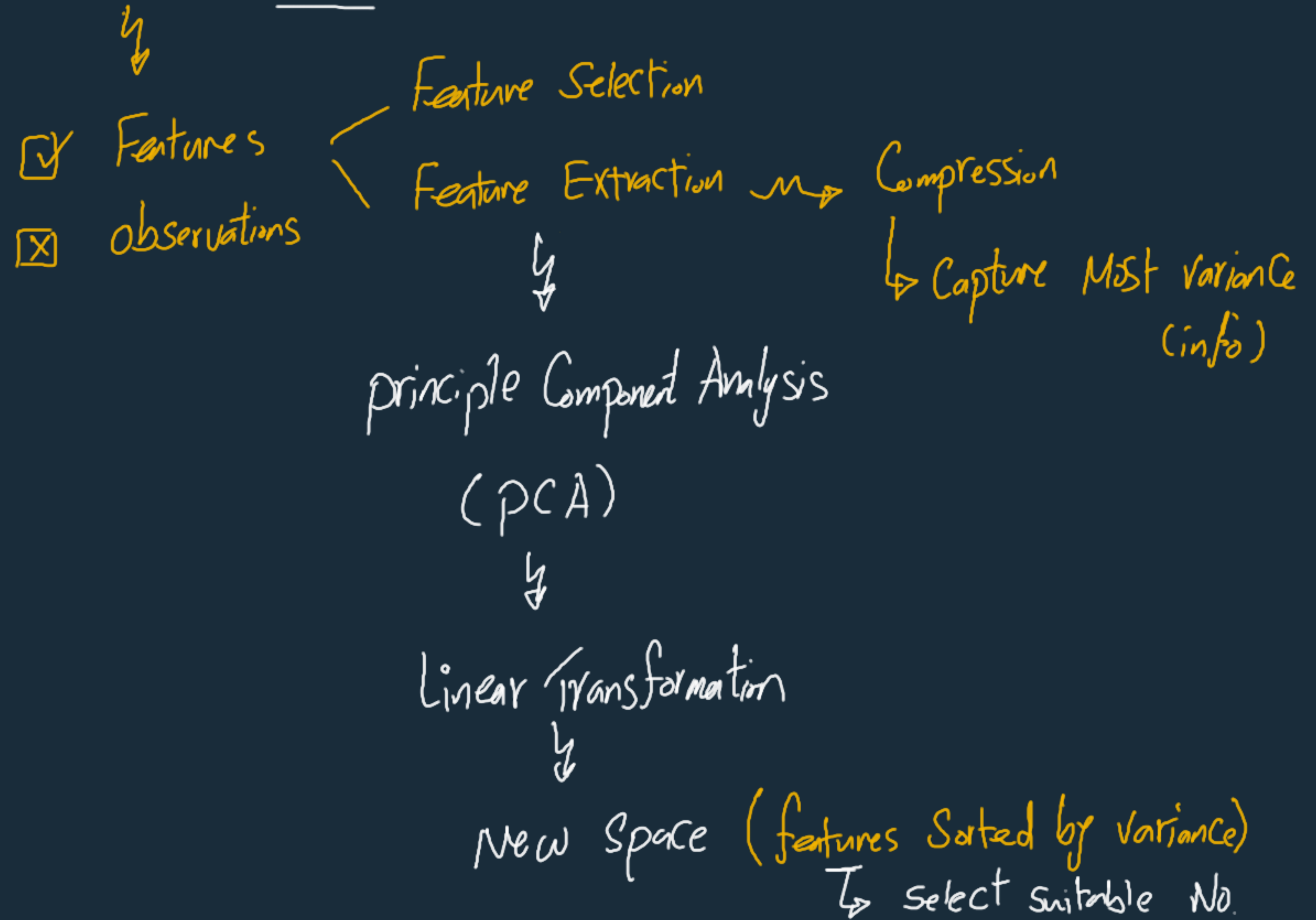
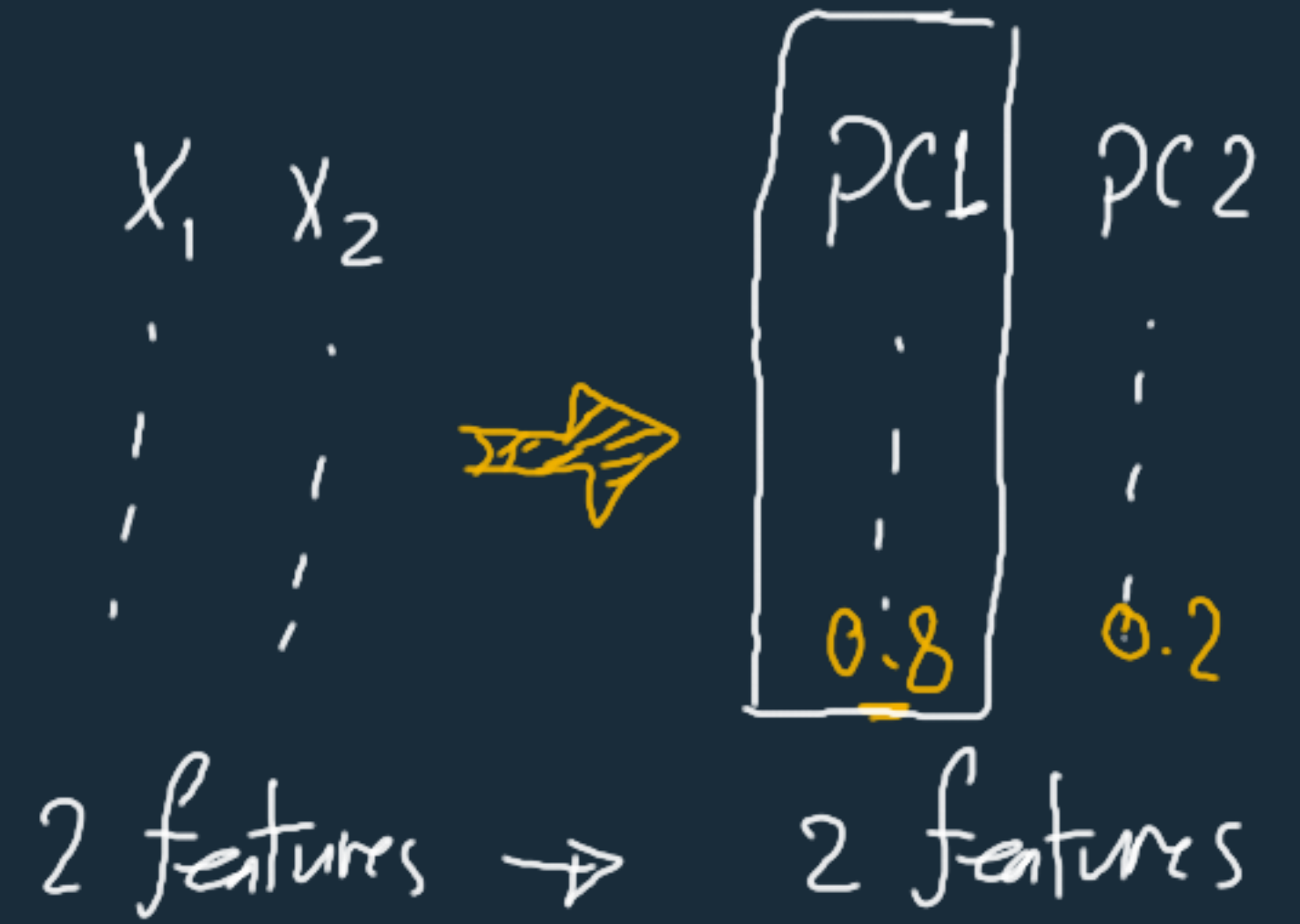
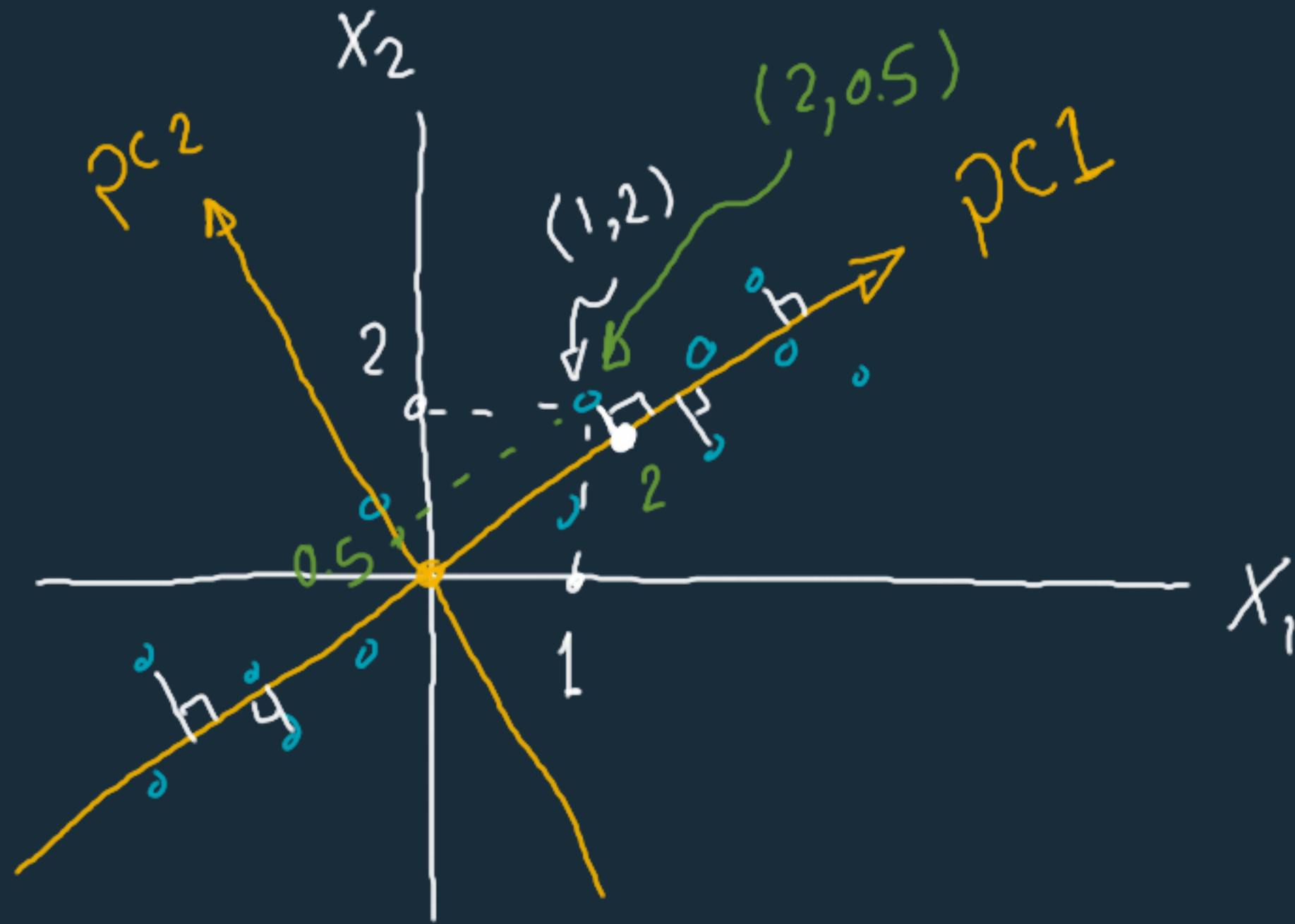


Dimension Reduction

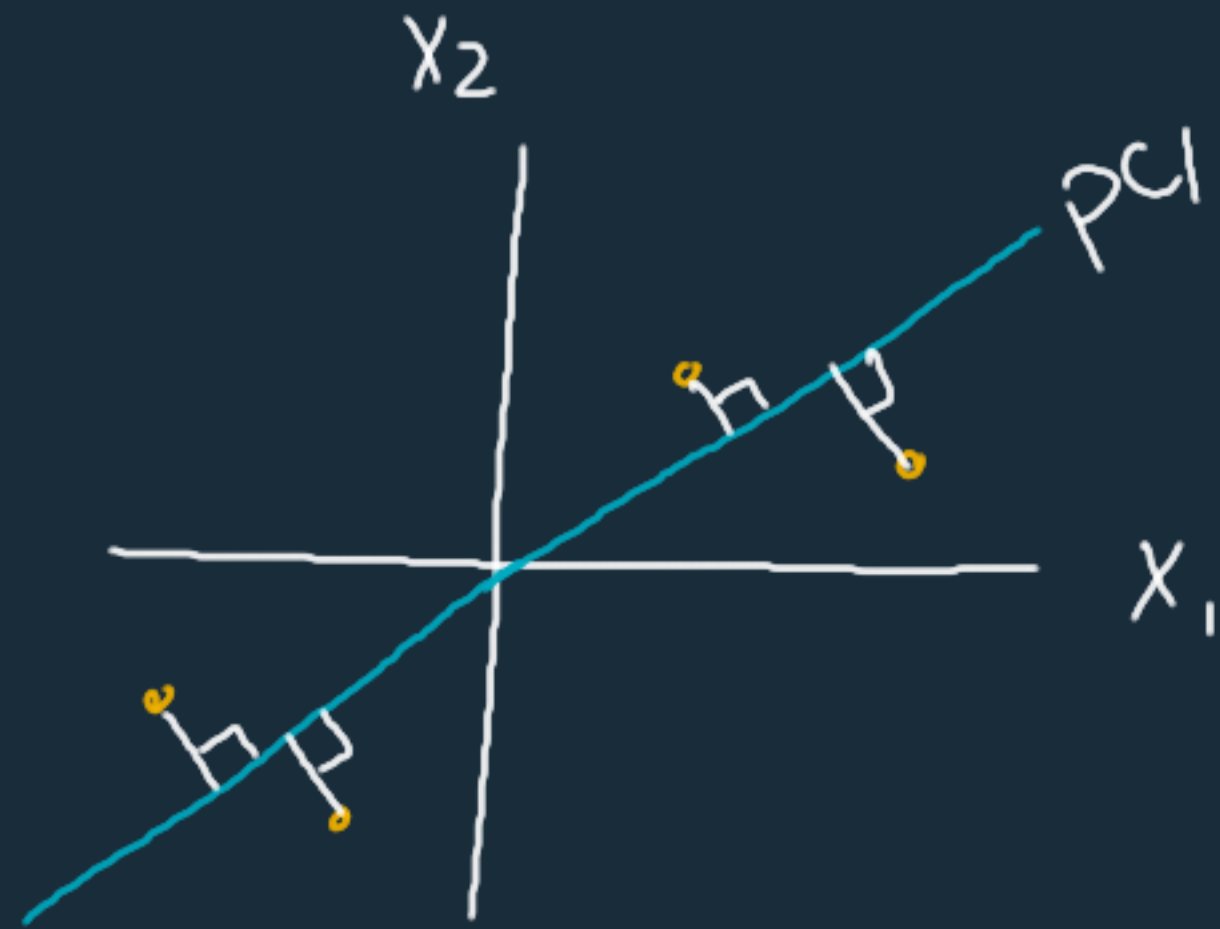




- ☐ How to get PC1 & PC2 ?
- ☐ (?, ?) on New Dimensions
- ☐ projection

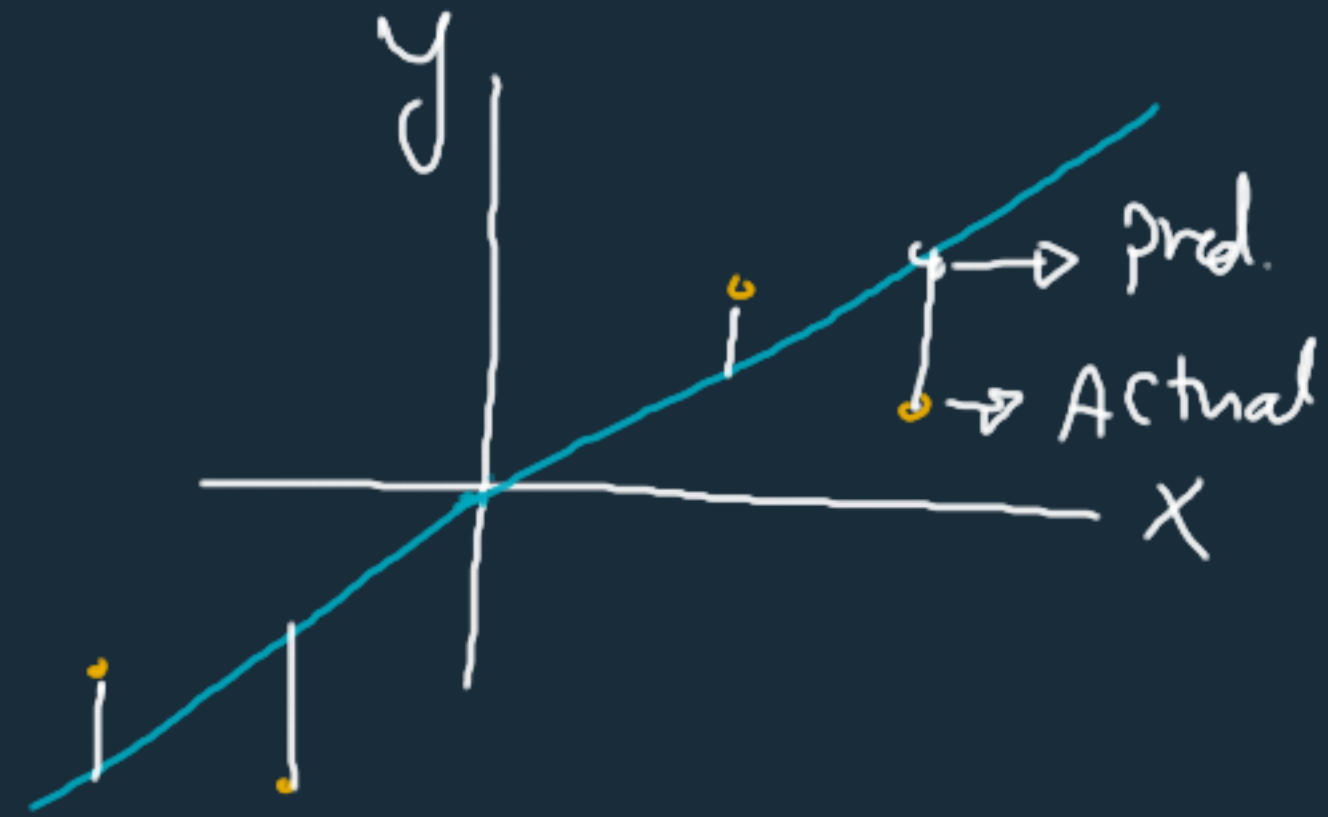
PC1 $\rightarrow$ 80% of var
 $\hookrightarrow$ Select PC1
 $\hookrightarrow$ drop PC2
 $\hookrightarrow$
 One feature

Unsupervised



Min. projection Error

Supervised



Min. prediction Error

① Calculate Covariance Matrix

$$\begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(x,y) & \text{var}(y) \end{bmatrix}_{n \times n} \rightsquigarrow n = \text{No. of features}$$

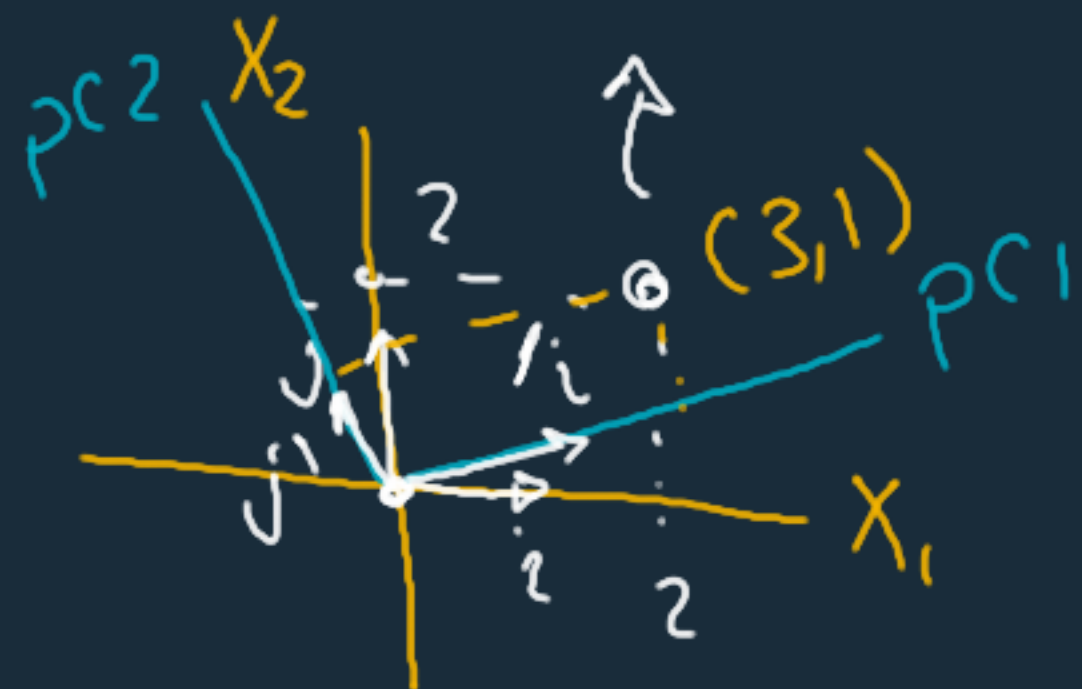
$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$[x_1, x_2] \cdot [i, j]$$

② Calculate Eigen Vectors & Eigen Values

(PC1, PC2) as unit vectors



③ Linear Transformation

Dot product

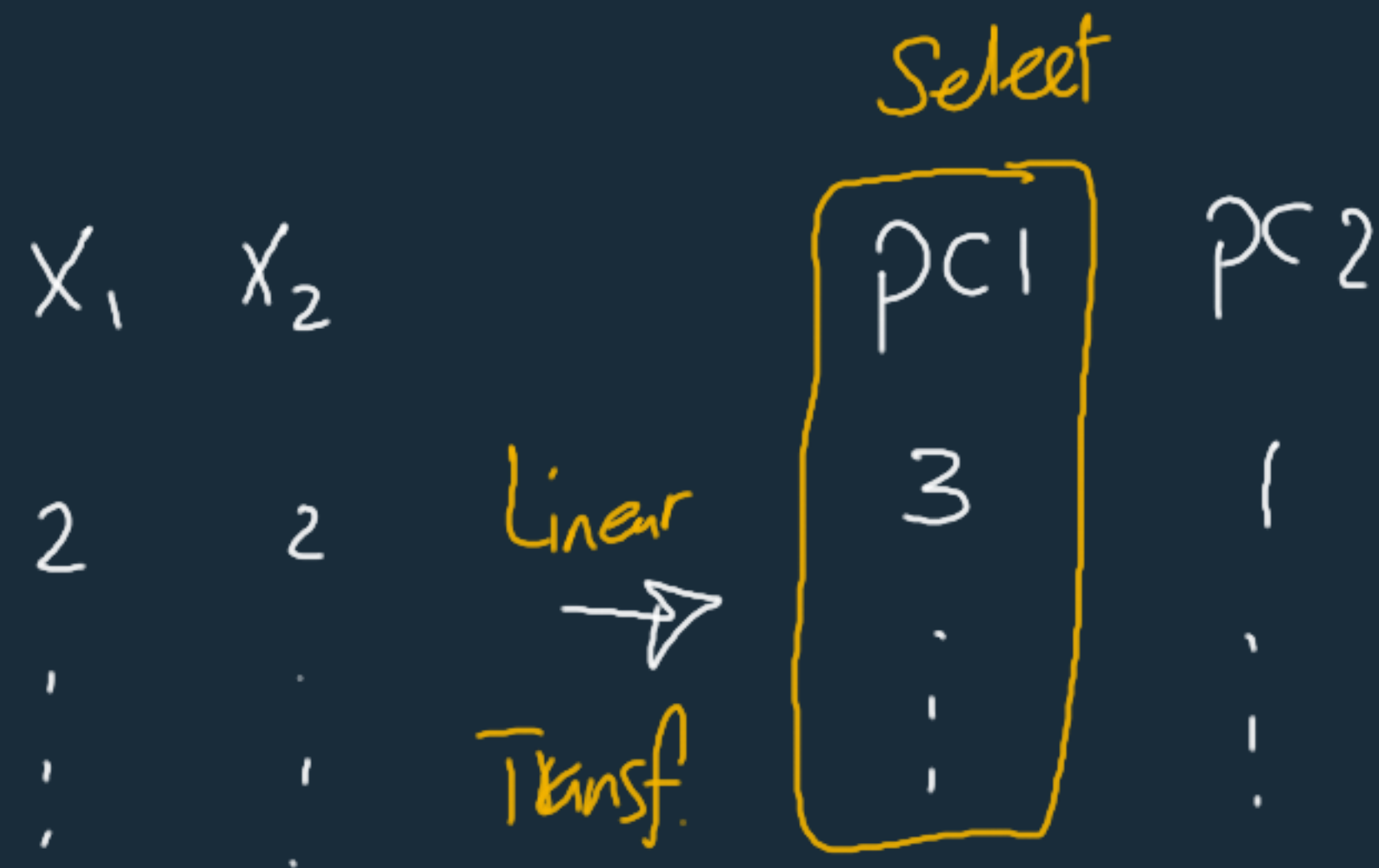
$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 \end{bmatrix} \text{ Most var.}$$

$$i' = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, j' = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$$

Eigen vectors

$$i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, j = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



2 features $\rightarrow$ 1 feature

$X_1 \quad X_2 \quad X_3 \quad \dots \quad X_n \rightarrow$

$\underbrace{\text{pc1} \quad \text{pc2} \quad \dots \quad \text{pcn}}$

$d < n$

Specific No of features
(2 or 3) $\rightarrow$ visualization

Min Variance
(0.8, 0.9, ...)

